

Figure 1. PHASE 11B experimental design

PART I Prospective 20-seed replication
2 directions x 2 models x 20 model seeds = 80 runs
each run persists source model AND scArches-adapted query model

Standardised per-run pipeline
1. train source 2. save source
3. query adapt 4. save adapted model
5. encode source+target with ADAPTED model

Recovered in PHASE 11B, missing in PHASE 11
target $q(z|x)$ posterior variance
 $U_{latent} = \text{mean}_j \sigma_{ij}^2$
from `get_latent_representation(return_dist=True)`

PART III Crossed design (Direction A only)
5 PHASE 10B source subsets x 5 model seeds x 2 models = 50 runs
separates source-data sampling from model-initialisation stochasticity

Model-seed
variability
(fixed source subset)

Source-subset
variability
(fixed model seed)

Interaction /
residual
(n=1 per cell)

PART IV-V Does full-source uncertainty predict BOTH instability sources?
Selective prediction stability across 20 seeds + cell-type retention imbalance